SS 1 PYTHON PROGRAMMING

TOPIC: LISTS IN PYTHON

WEEK 1: INTRODUCTION TO LISTS

Definitions

- **List**: A list is a data structure in Python used to store multiple values in a single variable.
- **Data Structure**: A way of organizing and storing data.
- **Element**: Each individual item inside a list.

Explanation

Lists are used when we want to store many related values together, such as names of students or scores.

Code Example

```
students = ["Ada", "John", "Musa", "Aisha"]

print(students)
```

Practical Project

Class Name Register: Create a list of 5 student names and display them.

WEEK 2: ACCESSING LIST ITEMS (INDEXING)

Definitions

- **Index**: The position of an item in a list.
- **Positive Index**: Index starting from 0.
- **Negative Index**: Index starting from -1 (last item).

Explanation

Python uses index numbers to access items in a list.

Code Example

```
students = ["Ada", "John", "Musa", "Aisha"]

print(students[0])

print(students[-1])
```

Practical Project

Attendance Checker: Display the first and last student in a list.

WEEK 3: LENGTH OF LIST & MEMBERSHIP

Definitions

- **len()**: A function used to count items in a list.
- **Membership Operator**: Used to check if an item exists in a list using in.

Code Example

```
students = ["Ada", "John", "Musa"]

print(len(students))

print("Ada" in students)
```

Practical Project

Exam Eligibility Checker: Count students and check if a name exists.

WEEK 4: ADDING ITEMS TO A LIST

Definitions

- **append()**: Adds an item to the end of a list.
- **insert()**: Adds an item at a specific position.

Code Example

```
students = ["Ada", "John"]

students.append("Musa")

students.insert(1, "Aisha")

print(students)
```

Practical Project

New Student Admission System: Add new students to a list.

WEEK 5: REMOVING ITEMS FROM A LIST

Definitions

- **remove()**: Removes a specific item.
- **pop()**: Removes an item using index.
- **del**: Deletes items or the entire list.

Code Example

```
students = ["Ada", "John", "Musa"]

students.remove("John")

students.pop()

print(students)
```

Practical Project

Student Withdrawal Record: Remove withdrawn students.

WEEK 6: UPDATING LIST ITEMS

Definitions

- **Mutable**: Data that can be changed.

Code Example

```
students = ["Ada", "Jon", "Musa"]

students[1] = "John"

print(students)
```

Practical Project

Correction System: Correct wrongly spelled names.

WEEK 7: LOOPING THROUGH LISTS

Definitions

- **Loop**: Repeats a block of code.

- **Iteration**: Going through items one by one.

Code Example

```
students = ["Ada", "John", "Musa"]

for name in students:
  print(name)
```

Practical Project

Class Roll Call: Print each student name.

WEEK 8: LIST SLICING

Definitions

- **Slicing**: Extracting a portion of a list.

Code Example

```
students = ["Ada", "John", "Musa", "Aisha", "Paul"]

print(students[0:3])
```

Practical Project

Group Assignment: Divide students into groups.

WEEK 9: SORTING & REVERSING LISTS

Definitions

- **sort()**: Arranges items in order.
- **reverse()**: Reverses the list order.

Code Example

```
students = ["Musa", "Ada", "John"]

students.sort()

students.reverse()
```

print(students)

Practical Project

Merit List Generator: Sort student names.

WEEK 10: LIST METHODS

Definitions

- **count()**: Counts occurrences of an item.
- **index()**: Returns index of an item.

Code Example

items = ["Pen", "Book", "Pen"]

print(items.count("Pen"))

print(items.index("Book"))

Practical Project

Class Inventory Manager: Track classroom items.

WEEK 11: NESTED LISTS

Definitions

- **Nested List**: A list inside another list.

Code Example

scores = [["Ada", 80, 90], ["John", 70, 85]]

print(scores[0][0], scores[0][1])

Practical Project

Student Score Sheet: Store multiple subject scores.

WEEK 12: LISTS WITH USER INPUT

Definitions

- **Input**: Data entered by user during program execution.

Code Example

```
students = []

name = input("Enter student name: ")

students.append(name)

print(students)
```

Practical Project

Interactive Registration System: Collect student names.

WEEK 13: FINAL MINI PROJECT

Project Title

Class Management System Using Lists

Requirements

- Add students
- Remove students
- Display list
- Sort students

Sample Code Skeleton

```
students = []
```